

Attitude Control of the Hydrobatic Intervention AUV *Cuttlefish* using Incremental Nonlinear Dynamic Inversion

Tom Slawik*, Shubham Vyas*, Leif Christensen*, Frank Kirchner*

Abstract—In this paper, we present an attitude control scheme for an autonomous underwater vehicle (AUV), which is based on incremental nonlinear dynamic inversion (INDI). Conventional model-based controllers depend on an exact model of the controlled system, which is difficult to find, especially for marine vehicles subject to highly nonlinear hydrodynamic effects. INDI trades off model accuracy with sensor accuracy by incorporating acceleration feedback and actuator output feedback to linearize a nonlinear system incrementally. Existing research primarily focuses on studying INDI on unmanned aerial vehicles. However, there is barely any research on controlling marine vehicles using INDI. The control task we are performing is a 90 degrees pitch-up maneuver, where the dual-arm intervention AUV *Cuttlefish* transitions from a horizontal traveling pose to a vertical intervention pose. We compare INDI to a classical model-based control scheme in the maritime test basin at DFKI RIC, Germany, and we find that INDI keeps the AUV much more steady both in the transitioning phase as well as in the station keeping phase.

I. INTRODUCTION

With the expansion of the blue economy, there has been a notable increase in subsea infrastructures. This includes a rise in aquaculture installations and offshore wind farms, along with their corresponding foundation structures, as documented by GWEC [1]. This is leading to a growing need for automated subsea operations not only within emerging markets but also within the traditional oil and gas industry. Human diving operations face many dangers and depend on good weather and have restricted depths [2]. Remotely Operated Vehicles (ROVs) are currently used for deeper dives, but using them requires large support vessels with special equipment, making ROV operations time-consuming and expensive. The new class of so-called *intervention* AUVs (I-AUVs) [3] aims to facilitate autonomous interaction with the subsea infrastructures mentioned above. The AUV *Cuttlefish*, shown in Figure 1 and described in detail in [4], is an I-AUV equipped with two arms and the capability for hydrobatic motions (see [5] on the term *hydrobatic*). Using its eight thrusters, it can take on arbitrary orientations in the water column to interact with features of an offshore structure that are otherwise difficult to reach. The vehicle can change its center of mass and center of buoyancy to

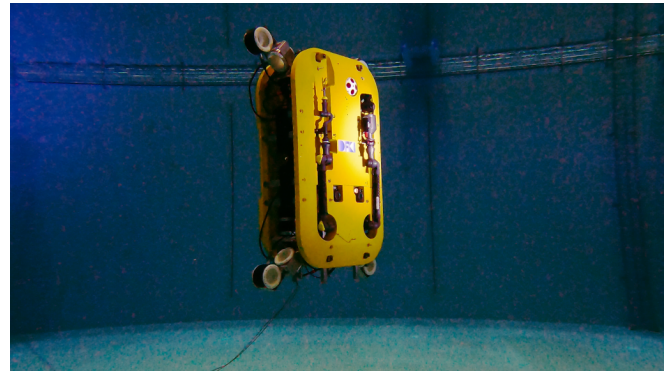


Fig. 1: AUV *Cuttlefish* in intervention pose in the test basin of DFKI RIC, Bremen, Germany.

achieve the desired stability for the current task, such as manipulation or transitioning into a new pose. The attitude control of autonomous underwater vehicles is crucial for their overall maneuverability and stability [6]. In this paper, we focus on the attitude control of the hydrobatic AUV *Cuttlefish* using incremental nonlinear dynamic inversion (INDI) [7]. This advanced control method allows for precise and agile maneuvering of the AUV, making it suitable for a wide range of underwater intervention tasks.

Due to *Cuttlefish*'s ability to assume an arbitrary pose, particularly an intervention pose with a pitch angle of 90 degrees, we do not rely on Euler angles for attitude control due to the Gimbal-Lock problem [8]. Instead, we use quaternions for parameterizing the orientation. During the transition to an upright pose, the AUV leaves its stable orientation, resulting in nonlinear restoring forces which must be compensated. Since the early 1990s, there has been a significant volume of academic research in literature focused on model-based control for underwater vehicles, the main results are summarized in [9] and [10]. In [11], a quaternion-based adaptive controller for a 6-DOF AUV is proposed, where the restoring forces are identified and canceled during operation. The resulting controller is evaluated in real world experiments on the AUV ODIN. Another work has been done by [12], where the center of buoyancy is identified using an Extended Kalman Filter (EKF) based on the Intervention-AUV SAUVIM. Besides hydrostatic restoring forces, there are hydrodynamic effects acting on the vehicle body during motion. Modeling these effects is difficult, since the shape of *Cuttlefish* is highly complex. Hydrodynamic effects which cannot be modeled also result in model uncertainty, which harms the control performance. An alternative to model-based control is sensor-based control, where model accuracy is traded off by sensor

An open-source implementation of the presented controller is available at https://github.com/dfki-ric-underactuated-lab/auv_control_indi. The accompanying video can be accessed at <https://youtu.be/8u8k607lpn4>

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使用增量非线性动态反演的水上干预自主水下航行器乌贼的姿态控制

汤姆 Slawik*, Shubham Vyas*, Leif Christensen*, Frank Kirchner*

Abstract—— 在本文中，我们提出了一种用于自主水下航行器（AUV）的姿态控制方案，该方案基于增量非线性动态反演（INDI）。传统的基于模型的控制依赖于受控系统的精确模型，而精确模型难以获得，尤其是对于受到高度非线性水动力效应影响的海洋车辆。INDI通过引入力反馈和执行器输出反馈，将非线性系统逐步线性化，从而在牺牲模型精度的情况下提高传感器精度。现有研究主要集中在研究无人飞行器上的INDI。然而，几乎没有关于使用INDI控制海洋车辆的研究。我们执行的控制任务是一个90度俯仰上仰机动，其中双臂干预AUV *Cuttlefish* 从水平航行姿态过渡到垂直干预姿态。我们在德国DFKI RIC的海事测试池中，将INDI与经典基于模型的控制方案进行比较，并发现

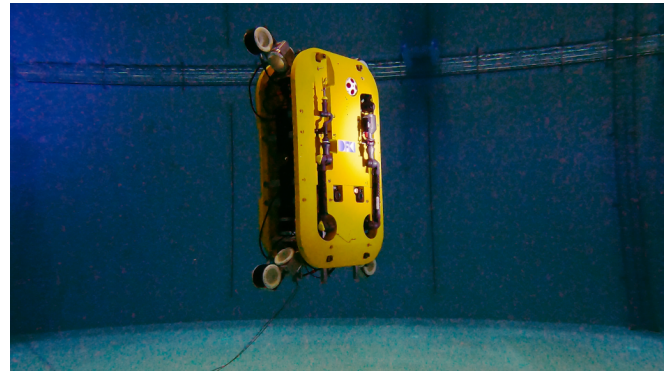


图1: AUV *Cuttlefish* 在德国不来梅 DFKI RIC 测试池中的干预姿态。

一、引言

随着蓝色经济的扩展，海底基础设施显著增加。这包括水产养殖设施和海上风电场的增加，以及它们相应的基础结构，正如GWEC [1] 所记录的那样。这导致了对海底自动化作业日益增长的需求，不仅在新兴市场中，而且在传统的石油和天然气行业中。人工潜水作业存在许多危险，依赖良好天气，并且潜水深度有限 [2]。遥控水下机器人（ROV）目前用于更深的潜水，但使用它们需要配备特殊设备的大型支援船，使ROV作业耗时且昂贵。新一代所谓的 *intervention* 自动水下机器人（AUVs）（I-AUVs）[3] 的目标是促进与上述海底基础设施的自主交互。如图1所示并在文献 [4] 中详细描述 of AUV *Cuttlefish* 是一款配备两条机械臂并具备水下运动能力的 I-AUV（关于参见 [5]）。

实现当前任务所需的稳定性，例如操作或转换到新的姿势。自主水下航行器的姿态控制对于其整体机动性和稳定性至关重要[6]。在本文中，我们重点研究使用增量非线性动态反演（INDI）[7]的水下机动AUV *Cuttlefish* 的姿态控制。这种先进的控制方法允许AUV进行精确而灵活的机动，使其适用于各种水下干预任务。

由于 *Cuttlefish* 能够摆出任意姿态，尤其是俯仰角为 90 度的干预姿态，我们在姿态控制中不依赖欧拉角，因为存在万向节锁问题 [8]。相反，我们使用四元数来参数化方向。在过渡到垂直姿态的过程中，AUV 会离开其稳定的姿态，从而产生必须补偿的非线性恢复力。自 1990 年代初以来，学术文献中有大量关于基于模型的水下航行器控制的研究，主要成果在 [9] 和 [10] 中进行了总结。在 [11] 中，提出了一种基于四元数的 6 自由度 AUV 自适应控制器，其中在操作过程中识别并抵消恢复力。所得到的控制器在 AUV ODIN 的实际实验进行了评估。另一项工作由 [12] 完成，其中利用扩展卡尔曼滤波器（EKF）基于干预 AUV S 来识别浮心。

An open-source implementation of the presented controller is available at https://github.com/dfki-ric-underactuated-lab/auv_control_indi. The accompanying video can be accessed at <https://youtu.be/8u8k607lpn4>

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accuracy. Incremental nonlinear dynamics inversion (INDI) is a sensor-based control scheme that iteratively linearizes a nonlinear system based on acceleration and input feedback. INDI has its origins in the unmanned aerial vehicle community [7], [13], [14]. However, there is barely any research on controlling marine vehicles using INDI. To date, the authors know about only one single publication [15], where the authors evaluate INDI to control an Unmanned Aerial-Underwater Vehicle, which is a quadrotor capable of diving. In this work, we present the first study of using INDI for motion control of a 6-DOF underwater vehicle. In Section II we will derive the mathematics behind INDI based on an underwater Fossen model [9]. Additionally, we will define the control laws for the model-based baseline controller, and the attitude and altitude control loops. Subsequently, in Section III we conduct real-world experiments of a control task, where the AUV *Cuttlefish* performs a 90-degrees pitch-up maneuver. Additionally, we perform a 6-DOF model identification of the AUV to compare INDI to a model-based controller. Our results show that INDI outperforms the model-based controller, especially in keeping the pose steady with minimum drift, which is crucial for intervention tasks.

II. IMPLEMENTATION

A. Vehicle Modeling

For modeling the AUV *Cuttlefish*, we use a 6-DOF Fossen model [9], where $\vec{v} = [u, v, w, p, q, r]^T \in \mathbb{R}^6$ is the body-fixed velocity vector, containing three linear velocities: surge, sway, and heave, and three angular velocities: roll, pitch, and yaw. $\vec{\eta} = [x, y, z, q_x, q_y, q_z, q_w]^T \in \mathbb{R}^7$ is the pose vector, where the orientation is encoded as a quaternion $\vec{q}$. The pose and velocity vectors are coupled through a nonlinear relationship:

$$\dot{\vec{\eta}} = J(\vec{\eta})\vec{v}, \quad (1)$$

where $J(\vec{\eta})$ is a Jacobian, also defined in [9]. The Fossen equations are given by:

$$M\ddot{\vec{v}} + C(\vec{v})\dot{\vec{v}} + \vec{d}(\vec{v}) + \vec{g}(\vec{\eta}) = \vec{\tau}, \quad (2)$$

where $\vec{v} \in \mathbb{R}^6$ is the acceleration vector, $M \in \mathbb{R}^{6 \times 6}$ is the mass-inertia matrix, and $\vec{\tau} \in \mathbb{R}^6$ is the input wrench. $C(\vec{v}) \in \mathbb{R}^{6 \times 6}$ is the coriolis-centripetal matrix, which is derived from the mass matrix M . The vector-field $\vec{d}(\vec{v})$ models the hydrodynamic drag. Due to its favorable trade-off between accuracy and robustness, we employ a quadratic drag model with $D_{\text{lin}}, D_{\text{quad}} \in \mathbb{R}^{6 \times 6}$ as parameters:

$$\vec{d}(\vec{v}) = D_{\text{lin}}\vec{v} + D_{\text{quad}}|\vec{v}|\vec{v} \in \mathbb{R}^6. \quad (3)$$

The function $\vec{g}(\vec{\eta}) \in \mathbb{R}^6$ models the hydrostatic forces which are caused by gravity and buoyancy, and is defined as:

$$\vec{g}(\vec{\eta}) = \Phi_g(\vec{\eta})\vec{\theta}_g, \quad (4)$$

where

$$\Phi_g(\vec{\eta}) = \begin{pmatrix} R(\vec{\eta})\vec{z} & 0_{3 \times 3} \\ \vec{0}_{3 \times 1} & S(R(\vec{\eta})\vec{z}) \end{pmatrix} \quad (5)$$

is a (6×4) matrix further described in [10]. Here, $\vec{z} = [0, 0, -1]^T$ is the axis of the gravitational field, and $\vec{\theta}_g \in \mathbb{R}^4$ is a parameter vector consisting of the net buoyancy force and three parameters for the restoring torques. $R(\vec{\eta})$ is the rotation matrix from world frame to body frame, and $S(\cdot)$ is the skew-symmetric cross-product operator.

The actuation $\vec{\tau}$ is modeled as a resulting wrench caused by the thrusters. It is calculated using the thruster allocation matrix $T \in \mathbb{R}^{6 \times 8}$ and the thruster RPM vector $u \in \mathbb{R}^8$:

$$\vec{\tau}(\vec{u}) = T f_{\text{thr}}(\vec{u}). \quad (6)$$

The relationship between the rotational speed u_i and its corresponding propulsive force is modeled as a quadratic curve $f_{\text{thr}}(u_i) = \kappa u_i |u_i|$. The coefficient κ was measured beforehand. For this actuation model, we ignore the moment of inertia caused by each individual thruster to simplify calculations.

B. Model Identification (Least-Squares)

We can express the dynamic equations given by (2) as a matrix-vector product:

$$\vec{\tau}_\theta(\vec{v}, \vec{\eta}) = \Phi(\vec{v}, \vec{\eta})\vec{\theta}, \quad (7)$$

where

$$\Phi(\vec{v}, \vec{\eta}) = [\vec{v}, \vec{v}, \vec{v}|\vec{v}|, \dots, \Phi_g(\vec{\eta})] \in \mathbb{R}^{6 \times p}. \quad (8)$$

$\vec{\theta}$ is a parameter vector containing 36 mass-matrix entries, 72 damping matrix entries, and 4 hydrostatic parameters:

$$\vec{\theta} = [X_{\dot{u}}, X_{\dot{v}}, \dots, X_u, X_{|u|u}, \dots, Y_{\dot{u}}, \dots]^T \in \mathbb{R}^p. \quad (9)$$

In order to identify the parameter vector $\vec{\theta}$, we collect $n+1$ samples from experiments and construct the regression matrix using (8):

$$A = \begin{bmatrix} \vec{v}_0, \vec{v}_0, \vec{v}_0|\vec{v}_0|, \dots, \Phi_g(\vec{\eta}_0) \\ \vdots \\ \vec{v}_n, \vec{v}_n, \vec{v}_n|\vec{v}_n|, \dots, \Phi_g(\vec{\eta}_n) \end{bmatrix} \in \mathbb{R}^{6n \times p}, \quad (10)$$

and the output vector

$$\vec{y} = [\vec{\tau}_0, \vec{\tau}_1, \dots, \vec{\tau}_n]^T \in \mathbb{R}^{6n}. \quad (11)$$

We want to minimize the prediction error using the sum of squared errors $J(\vec{\theta}) = \|\vec{y} - \vec{\tau}_\theta(\vec{v}, \vec{\eta})\|_2$. A solution is given by the Moore-Penrose pseudoinverse [16]:

$$\vec{\hat{\theta}} = (A^T A)^{-1} A^T \vec{y}. \quad (12)$$

C. Feedback Linearization (FBL)

State feedback linearization (also known as „Nonlinear Dynamic Inversion“ (NDI) in aerospace control systems) is a technique to transform a nonlinear system into a linear system through a feedback loop [17]. Recall the dynamics equation presented in (2). We rewrite this equation as follows:

$$M\ddot{\vec{v}} + \vec{f}(\vec{v}, \vec{\eta}) = \vec{\tau}, \quad (13)$$

精度。增量非线性动力学反演 (INDI) 是一种基于传感器的控制方案，它基于加速度和输入反馈迭代地对非线性系统进行线性化。INDI 起源于无人机社区 [7], [13], [14]。然而，几乎没有关于使用 INDI 控制海洋车辆的研究。到目前为止，作者仅知道有一篇相关的出版物 [15]，文中作者评估了 INDI 控制一种无人水下飞行器 (AUV)，这是一种能够潜水的四旋翼。在这项工作中，我们首次研究了使用 INDI 对 6 自由度水下车辆进行运动控制。在第二节中，我们将基于水下 Fossen 模型 [9] 推导 INDI 背后的数学原理。此外，我们将定义基于模型的基线控制器的控制律，以及姿态和高度控制环路。随后，在第三节中，我们进行了一个控制任务的实际实验，其中自主水下航行器 (AUV) *Cuttlefish* 执行了一个 90° 俯仰上升动作

二、实施

A. Vehicle Modeling

为了对 AUV *Cuttlefish* 进行建模，我们使用了一个 6 自由度 Fossen 模型 [9]，其中 $\vec{v} = [u, v, w, p, q, r]^T \in \mathbb{R}^6$ 是机体固定的速度向量，包含三个线速度：纵向、横向和升沉速度，以及三个角速度：滚转、俯仰和偏航。 $\vec{\eta} = [x, y, z, q_x, q_y, q_z, q_w]^T \in \mathbb{R}^7$ 是位姿向量，其中方向以四元数 $\vec{q}$ 编码。位姿向量和速度向量通过非线性关系耦合：

$$\dot{\vec{\eta}} = J(\vec{\eta})\vec{v}, \quad (1)$$

其中 $J(\vec{\eta})$ 是雅可比矩阵，也在 [9] 中定义。Fossen 方程如下所示：

$$M\ddot{\vec{v}} + C(\vec{v})\dot{\vec{v}} + \vec{d}(\vec{v}) + \vec{g}(\vec{\eta}) = \vec{\tau}, \quad (2)$$

其中 $\vec{v} \in \mathbb{R}^6$ 是加速度向量， $M \in \mathbb{R}^{6 \times 6}$ 是质量-惯性矩阵， $\vec{\tau} \in \mathbb{R}^6$ 是输入力矩。 $C(\vec{v}) \in \mathbb{R}^{6 \times 6}$ 是科里奥利-向心矩阵，由质量矩阵 M 导出。向量场 $\vec{d}(\vec{v})$ 模拟流体阻力。由于其在精度和鲁棒性之间的良好折衷，我们采用二次阻力模型，其中 $D_{\text{lin}}, D_{\text{quad}} \in \mathbb{R}^{6 \times 6}$ 为参数：

$$\vec{d}(\vec{v}) = D_{\text{lin}}\vec{v} + D_{\text{quad}}|\vec{v}|\vec{v} \in \mathbb{R}^6. \quad (3)$$

函数 $\vec{g}(\vec{\eta}) \in \mathbb{R}^6$ 模拟由重力和浮力引起的静水力，并定义为：

$$\vec{g}(\vec{\eta}) = \Phi_g(\vec{\eta})\vec{\theta}_g, \quad (4)$$

哪里

$$\Phi_g(\vec{\eta}) = \begin{pmatrix} R(\vec{\eta})\vec{z} & 0_{3 \times 3} \\ \vec{0}_{3 \times 1} & S(R(\vec{\eta})\vec{z}) \end{pmatrix} \quad (5)$$

是一个 (6×4) 矩阵，进一步描述见 [10]。这里， $\vec{z} = [0, 0, -1]^T$ 是重力场的轴，而 $\vec{\theta}_g \in \mathbb{R}^4$ 是由净浮力和三个恢复力矩参数组成的参数向量。 $R(\vec{\eta})$ 是从世界坐标系到机体坐标系的旋转矩阵， $S(\cdot)$ 是斜对称的叉乘运算符。

驱动 $\vec{\tau}$ 被建模为由推进器造成的结果力矩。它通过推进器分配矩阵 $T \in \mathbb{R}^{6 \times 8}$ 和推进器转速向量 $u \in \mathbb{R}^8$ 计算得出：

$$\vec{\tau}(\vec{u}) = T f_{\text{thr}}(\vec{u}). \quad (6)$$

旋转速度 u_i 与其对应的推进力的关系被建模为二次曲线 $f_{\text{thr}}(u_i) = \kappa u_i |u_i|$ 。系数 κ 已提前测量。对于该驱动模型，我们忽略了每个单独推进器所产生的转动惯量，以简化计算。

B. Model Identification (Least-Squares)

我们可以将由 (2) 给出的动力学方程表示为矩阵-向量乘积：

$$\vec{\tau}_\theta(\vec{v}, \vec{\eta}) = \Phi(\vec{v}, \vec{\eta})\vec{\theta}, \quad (7)$$

哪里

$$\Phi(\vec{v}, \vec{\eta}) = [\vec{v}, \vec{v}, \vec{v}|\vec{v}|, \dots, \Phi_g(\vec{\eta})] \in \mathbb{R}^{6 \times p}. \quad (8)$$

$\vec{\theta}$ 是一个包含 36 个质量矩阵条目、72 个阻尼矩阵条目和 4 个静水参数的参数向量：

$$\vec{\theta} = [X_{\dot{u}}, X_{\dot{v}}, \dots, X_u, X_{|u|u}, \dots, Y_{\dot{u}}, \dots]^T \in \mathbb{R}^p. \quad (9)$$

为了识别参数向量 $\vec{\theta}$ ，我们从实验中收集 $n + 1$ 个样本，并使用 (8) 构建回归矩阵：

$$A = \begin{bmatrix} \vec{v}_0, \vec{v}_0, \vec{v}_0|\vec{v}_0|, \dots, \Phi_g(\vec{\eta}_0) \\ \vdots \\ \vec{v}_n, \vec{v}_n, \vec{v}_n|\vec{v}_n|, \dots, \Phi_g(\vec{\eta}_n) \end{bmatrix} \in \mathbb{R}^{6n \times p}, \quad (10)$$

以及输出向量

$$\vec{y} = [\vec{\tau}_0, \vec{\tau}_1, \dots, \vec{\tau}_n]^T \in \mathbb{R}^{6n}. \quad (11)$$

我们希望使用平方误差和 $J(\vec{\theta}) = \|\vec{y} - \vec{\tau}_\theta(\vec{v}, \vec{\eta})\|_2$ 来最小化预测误差。一个解由 Moore-Penrose 伪逆 [16] 给出：

$$\vec{\hat{\theta}} = (A^T A)^{-1} A^T \vec{y}. \quad (12)$$

C. Feedback Linearization (FBL)

状态反馈线性化（在航空航天控制系统中也称为“非线性动态逆” (NDI)）是一种通过反馈回路将非线性系统转换为线性系统的技术 [17]。回顾在 (2) 中给出的动力学方程。我们将此方程重写如下：

$$M\ddot{\vec{v}} + \vec{f}(\vec{v}, \vec{\eta}) = \vec{\tau}, \quad (13)$$

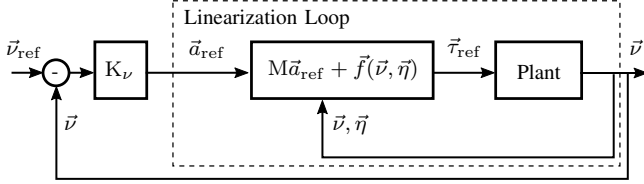


Fig. 2: Block diagram of the FBL-based velocity control loop

where $\vec{f}(\vec{v}, \vec{\eta})$ is a smooth vector field describing the nonlinear system dynamics:

$$\vec{f}(\vec{v}, \vec{\eta}) = C(\vec{v})\vec{v} + \vec{d}(\vec{v}) + \vec{g}(\vec{\eta}) . \quad (14)$$

Now, we introduce a new control variable $\vec{\nu} = \vec{a}_{ref}$, which acts as the input vector for the linearized system:

$$\vec{\tau}_{ref} = M\vec{a}_{ref} + \vec{f}(\vec{v}, \vec{\eta}) , \quad (15)$$

where $\vec{\tau}_{ref}$ is the commanded wrench. Feedback linearization not only cancels the modeled nonlinearities, it also decouples each input $\vec{a}_{ref,i}$ to only affect its corresponding output $\vec{v}_i$. However, the approach follows the assumptions that a) an exact model of the system dynamics is given, and b), that the state vector is exactly known at each time step.

A block diagram showing feedback linearization in conjunction with the velocity controller is shown in Figure 2.

D. Incremental Nonlinear Dynamic Inversion (INDI)

Incremental Nonlinear Dynamic Inversion (INDI) is an extension of Nonlinear Dynamic Inversion (NDI) / Feedback Linearization. The idea is to incrementally update the control command $\vec{\tau}_{ref}$ in order to linearize the system. At each time step, the increment is calculated from accelerometer readings and measurements of the actual thruster rates. This is opposed to NDI, where the control command is calculated directly from the model.

To derive the control law for INDI, we first rearrange the Fossen equation (13) to an implicit form:

$$\vec{\nu} = M^{-1} [\vec{\tau} - \vec{f}(\vec{v}, \vec{\eta})] . \quad (16)$$

Now, we develop the first-order Taylor expansion of the previous equation, where $\vec{v}_0, \vec{\eta}_0$ and $\vec{\tau}_0$ correspond to the measured state and input vectors:

$$\begin{aligned} \vec{\nu} \approx & M^{-1} [\vec{\tau}_0 - \vec{f}(\vec{v}_0, \vec{\eta}_0)] \\ & + \frac{\partial}{\partial \vec{\tau}} [M^{-1} \vec{\tau}]_{\vec{\tau}=\vec{\tau}_0} (\vec{\tau} - \vec{\tau}_0) \\ & + \frac{\partial}{\partial \vec{v}} [M^{-1} \vec{f}(\vec{v}, \vec{\eta})]_{\vec{v}=\vec{v}_0} (\vec{v} - \vec{v}_0) \\ & + \frac{\partial}{\partial \vec{\eta}} [M^{-1} \vec{f}(\vec{v}, \vec{\eta})]_{\vec{\eta}=\vec{\eta}_0} (\vec{\eta} - \vec{\eta}_0) . \end{aligned} \quad (17)$$

We make the assumption that the thruster dynamics are much faster than the dynamics of the AUV body:

$$|\vec{f}(\vec{v}, \vec{\eta}) - \vec{f}(\vec{v}_0, \vec{\eta}_0)| \ll |\vec{\tau} - \vec{\tau}_0| . \quad (18)$$

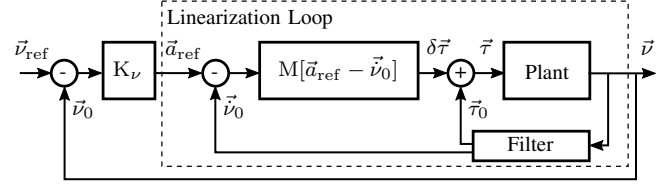


Fig. 3: Block diagram of the INDI-based velocity controller

With this assumption, we simplify the dynamics to:

$$\begin{aligned} \vec{\nu} \approx & M^{-1} [\vec{\tau}_0 - \vec{f}(\vec{v}_0, \vec{\eta}_0)] \\ & + M^{-1} (\vec{\tau} - \vec{\tau}_0) . \end{aligned} \quad (19)$$

Since $\vec{\nu}_0 = M^{-1} [\vec{\tau}_0 - \vec{f}(\vec{v}_0, \vec{\eta}_0)]$, we replace the dynamic terms of the previous equation with the measured acceleration $\vec{\nu}_0$:

$$\vec{\nu} \approx \vec{\nu}_0 + M^{-1} (\vec{\tau} - \vec{\tau}_0) . \quad (20)$$

Inverting this equation leads to

$$\delta \tau = \vec{\tau} - \vec{\tau}_0 = M [\vec{\nu} - \vec{\nu}_0] , \quad (21)$$

which is the increment of the control wrench for one time step. If we introduce $\vec{\nu} = \vec{a}_{ref}$ to be our new control variable (as in feedback linearization), we finally get the incremental control law:

$$\vec{\tau}_{ref} = M [\vec{a}_{ref} - \vec{\nu}_0] + \vec{\tau}_0 . \quad (22)$$

Note that $\vec{\tau}_0$ is the actual input wrench, which is obtained by the actual thruster's rpm readings and the actuation model described in (6). As the angular acceleration $\vec{\omega} \in \mathbb{R}^3$ cannot be measured directly, we estimate it using finite differences: $\vec{\omega} \approx (\vec{\omega}_{t+1} - \vec{\omega}_t)/dt$, where $dt = 0.01$ s is the sampling interval. To reduce noise, we apply a first-order Butterworth low-pass filter with a cutoff frequency of 4.0 Hz to the angular velocity readings before differentiation. The same low-pass filter is also applied to the thruster rate readings $\vec{u}_0$ and the linear acceleration, since the phase must be synchronized in order to avoid oscillations. An overview of the INDI linearization loop is shown in Figure 3.

E. Velocity Controller (Inner Loop)

The linearized system can now be controlled using linear control theory. For controlling the linear and angular velocities we use a proportional control law:

$$\vec{a}_{ref} = K_v (\vec{v}_{ref} - \vec{v}) , \quad (23)$$

where $\vec{v}_{ref} \in \mathbb{R}^6$ is the desired velocity vector. To find the optimal gain parameters, we use trial-and-error to find a diagonal gain matrix which is optimized such that the step response of the maximum velocity control commands does not exceed the force/torque limits of 300 N / 300 Nm (which are given due to internal safety and power regulations), while maintaining an optimal tracking performance:

$$K_v = \text{diag}(0.3, 0.3, 0.3, 1.0, 0.7, 0.7) . \quad (24)$$

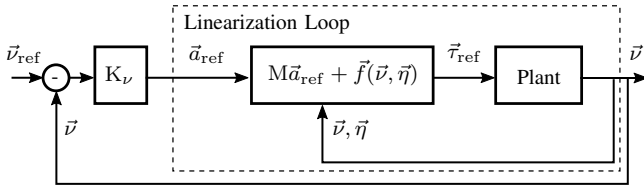


图2: 基于FBL的速度控制回路方框图

其中 $\vec{f}(\vec{v}, \vec{\eta})$ 是描述非线性系统动力学的光滑向量场:

$$\vec{f}(\vec{v}, \vec{\eta}) = C(\vec{v})\vec{v} + \vec{d}(\vec{v}) + \vec{g}(\vec{\eta}). \quad (14)$$

现在, 我们引入一个新的控制变量 $\vec{v} = \vec{a}_{ref}$, 它作为线性化系统的输入向量:

$$\vec{\tau}_{ref} = M\vec{a}_{ref} + \vec{f}(\vec{v}, \vec{\eta}), \quad (15)$$

其中 $\vec{\tau}_{ref}$ 是指令力矩。反馈线性化不仅可以抵消建模的非线性, 还可以使每个输入 $\vec{a}_{ref,i}$ 与其对应的输出 $\vec{v}_i$ 解耦。然而, 该方法遵循以下假设: a) 已给出系统动力学的精确模型, b) 在每个时间步上状态向量是完全已知的。

图 2 显示了与速度控制器结合使用的反馈线性化的框图。

D. Incremental Nonlinear Dynamic Inversion (INDI)

增量非线性动态反演 (INDI) 是非线性动态反演 (NDI) / 反馈线性化的一个扩展。其思想是增量更新控制指令 $\vec{\tau}_{ref}$ 以实现系统线性化。在每个时间步, 增量通过加速度计读数和实际推进器速率的测量来计算。这与 NDI 相反, NDI 是直接来自模型计算控制指令。

为了推导 INDI 的控制律, 我们首先将 Fossen 方程 (13) 重新排列为隐式形式:

$$\vec{v} = M^{-1} [\vec{\tau} - \vec{f}(\vec{v}, \vec{\eta})]. \quad (16)$$

现在, 我们对前一个方程展开一阶泰勒展开, 其中 $\vec{v}_0, \vec{\eta}_0$ 和 $\vec{\tau}_0$ 对应于测量的状态向量和输入向量:

$$\begin{aligned} \vec{v} \approx & M^{-1} [\vec{\tau}_0 - \vec{f}(\vec{v}_0, \vec{\eta}_0)] \\ & + \frac{\partial}{\partial \vec{\tau}} [M^{-1} \vec{\tau}]_{\vec{\tau}=\vec{\tau}_0} (\vec{\tau} - \vec{\tau}_0) \\ & + \frac{\partial}{\partial \vec{v}} [M^{-1} \vec{f}(\vec{v}, \vec{\eta})]_{\vec{v}=\vec{v}_0} (\vec{v} - \vec{v}_0) \\ & + \frac{\partial}{\partial \vec{\eta}} [M^{-1} \vec{f}(\vec{v}, \vec{\eta})]_{\vec{\eta}=\vec{\eta}_0} (\vec{\eta} - \vec{\eta}_0). \end{aligned} \quad (17)$$

我们假设推进器的动力学远快于 AUV 本体的动力学:

$$|\vec{f}(\vec{v}, \vec{\eta}) - \vec{f}(\vec{v}_0, \vec{\eta}_0)| \ll |\vec{\tau} - \vec{\tau}_0|. \quad (18)$$

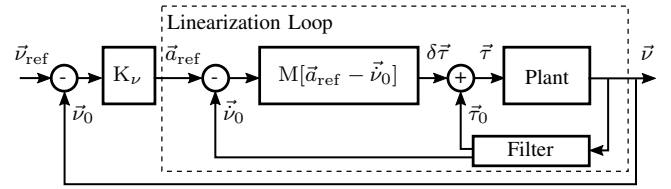


图3: 基于INDI的速度控制器框图

在这个假设下, 我们将动力学简化为:

$$\begin{aligned} \vec{v} \approx & M^{-1} [\vec{\tau}_0 - \vec{f}(\vec{v}_0, \vec{\eta}_0)] \\ & + M^{-1} (\vec{\tau} - \vec{\tau}_0). \end{aligned} \quad (19)$$

自从 $\vec{v}_0 = M^{-1} [\vec{\tau}_0 - \vec{f}(\vec{v}_0, \vec{\eta}_0)]$, 我们将前一方程的动态项替换为测得的加速度 $\vec{v}_0$:

$$\vec{v} \approx \vec{v}_0 + M^{-1} (\vec{\tau} - \vec{\tau}_0). \quad (20)$$

将这个方程反过来得到

$$\delta \tau = \vec{\tau} - \vec{\tau}_0 = M [\vec{v} - \vec{v}_0], \quad (21)$$

它是一个时间步长的控制扳手增量。如果我们引入 $\vec{v} = \vec{a}_{ref}$ 作为新的控制变量 (如在反馈线性化中), 我们最终得到增量控制律:

$$\vec{\tau}_{ref} = M [\vec{a}_{ref} - \vec{v}_0] + \vec{\tau}_0. \quad (22)$$

请注意, $\vec{\tau}_0$ 是实际的输入力矩, 它是通过实际推进器的转速读数以及在(6)中描述的执行模型获得的。由于角加速度 $\vec{\omega} \in \mathbb{R}^3$ 无法直接测量, 我们使用有限差分来估计它: $\vec{\omega} \approx (\vec{\omega}_{t+1} - \vec{\omega}_t)/dt$, 其中 $dt = 0.01$ s 是采样间隔。为了降低噪声, 我们在对角速度读数求导之前, 应用截止频率为 4.0 Hz 的一阶巴特沃斯低通滤波器。同样的低通滤波器也应用于推进器转速读数 $\vec{u}_0$ 和线性加速度, 因为必须同步相位以避免振荡。INDI 线性化回路的概述如图 3 所示。

E. Velocity Controller (Inner Loop)

现在可以使用线性控制理论来控制线性化系统。为了控制线性和角速度, 我们使用比例控制律:

$$\vec{a}_{ref} = K_v (\vec{v}_{ref} - \vec{v}), \quad (23)$$

$\vec{v}_{ref} \in \mathbb{R}^6$ 是期望的速度向量。为了找到最优的增益参数, 我们使用反复试验的方法来找到一个对角增益矩阵, 该矩阵经过优化, 使得最大速度控制指令的阶跃响应不会超过 300 N / 300 Nm 的力/扭矩限制 (这是由于内部安全和功率规定所给定的), 同时保持最佳的跟踪性能:

$$K_v = \text{diag}(0.3, 0.3, 0.3, 1.0, 0.7, 0.7). \quad (24)$$

F. Attitude Controller (Outer Loop)

Euler angles are not suitable for describing the orientation of an AUV that is designed for variable attitude maneuvers, since certain configurations cause a loss of one degree of freedom. This problem is also called the „Gimbal-Lock“ problem [8]. For this reason, we use quaternions to describe the orientation of the AUV. To control the orientation, we implement an attitude controller which is derived from the work of Chaturvedi et al. [8]:

$$\omega_{\text{ref}}(\mathbf{R}, \mathbf{R}_d) = \mathbf{K}_\Omega \sum_{i=1}^3 \mathbf{e}_i \times (\mathbf{R}_d^\top \mathbf{R}^\top \mathbf{e}_i) . \quad (25)$$

The attitude controller generates an angular velocity command $\omega_{\text{ref}} \in \mathbb{R}^3$ based on rotation matrices $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ (current orientation), and $\mathbf{R}_d \in \mathbb{R}^{3 \times 3}$ (desired orientation). Both matrices can be directly calculated from the current and desired quaternions. $[\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3]$ is the identity matrix. $\mathbf{K}_\Omega$ is a proportional gain matrix, whose elements are chosen during real-world experiments such that there is minimum overshoot during a pitch-up maneuver:

$$\mathbf{K}_\Omega = \text{diag}(0.15, 0.15, 0.15) . \quad (26)$$

To avoid too large angular velocities, $\vec{\omega}_{\text{ref}}$ is bounded to $\|\vec{\omega}_{\text{ref}}\|_{\text{max}} < 0.2 \text{ rad/s}$.

G. Depth Controller

In order to keep the depth of the AUV steady, a proportional-gain depth controller is implemented:

$$\vec{v}_{\text{ref}} = \mathbf{K}_{\text{pos}} \mathbf{R}_B^I [[0, 0, z_{\text{ref}}]^\top - [0, 0, z]^\top] , \quad (27)$$

where z_{ref} is the commanded depth, and z is the actual, measured depth. $\vec{v}_{\text{ref}} \in \mathbb{R}^3$ is the commanded linear velocity, and $\mathbf{R}_B^I$ is a rotation matrix from world frame to body frame. $\mathbf{K}_\Omega$ is a proportional gain matrix, whose elements are chosen during real-world experiments to result in minimum overshoot:

$$\mathbf{K}_{\text{pos}} = \text{diag}(0.5, 0.5, 0.5) . \quad (28)$$

The maximum linear velocity command $\vec{v}_{\text{ref}}$ is bounded to $\pm[0.5 \text{ m/s}, 0.5 \text{ m/s}, 0.2 \text{ m/s}]$.

III. EVALUATION

A. Experimental Setup

TABLE I: *Cuttlefish* Specifications [4]

Specification	Value
Length / Width / Height	2.8 m / 2.0 m / 0.8 m
Weight	1200 kg
Inertial Navigation System	iXblue Phins Compact C3
DVL	Rowe SeaPilot
Pressure Sensor	Keller PAA-33x
Rated Driving Power	21.6 kW (8x 2.7 kW)

The experiments are conducted in the maritime test basin of DFKI RIC, Bremen, Germany¹. The target vehicle is the

¹<https://robotik.dfk-bremen.de/en/research/research-facilities-labs/maritime-infrastructure>

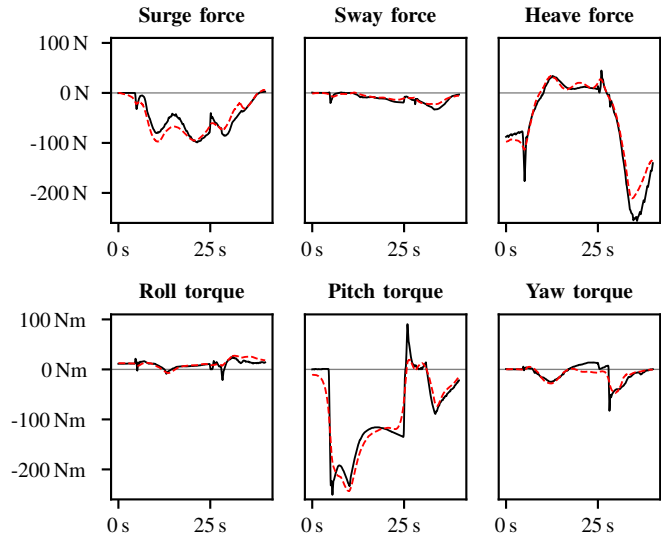


Fig. 4: Model identification test (dashed line: prediction)

dual-arm intervention AUV *Cuttlefish* [4], which is based on a steel frame construction equipped with 13 pressure housings for electronics and PVC-based foam, to make the vehicle positive buoyant. The two robotic arms are not used for our experiments and are fixed to the body. To control the 1200 kg heavy vehicle, eight 2.7 kW rim-driven thrusters are installed (four vertical ones and four horizontal ones). The thrusters were developed by Wittenstein cyber motor GmbH specifically for *Cuttlefish*, and are designed to be precisely controllable, even at very low RPM. The measured time constant of the thruster acceleration is $T_{\text{thr}} = 0.672 \text{ s}$. For estimating the attitude, velocities and linear accelerations, we use the iXblue Phins Compact C3 inertial navigation system. The Phins C3 is based on a fiber-optic gyroscope, offering high-precision rotation rate measurements with a resolution of 0.001 deg/s. Additionally, the Phins C3 is capable of estimating the orientation in roll and pitch with an accuracy of 0.05 deg RMS. The accuracy of the linear velocity estimate is 0.1 kn = 0.0514 m/s. The Phins C3 uses its own Kalman filter implementation to fuse data with various external sensors, in our setup a Rowe SeaPilot DVL to measure the speed over ground, and the Keller PAA-33x pressure sensor. A summary of the basic specifications is given in Table I. *Cuttlefish* follows a centralized computer architecture, where all software modules are running on a single x86-based computer, equipped with an Intel(R) Core(TM) i7-8700 CPU and 32 GB of RAM. The software stack is composed of Ubuntu 18.04 as the host system, running a ROS Noetic environment in a Docker container. Our control stack is split into two ROS-nodes, where one ROS-node is responsible for high-level tracking of depth and attitude, while the other ROS-node embeds the low-level FBL- and INDI - based velocity controllers. The update rate of the control stack amounts to 100 Hz.

F. Attitude Controller (Outer Loop)

欧拉角不适合描述设计用于可变姿态机动的AUV的姿态，因为某些构型会导致一个自由度的丧失。这个问题也被称为“万向锁”问题[8]。因此，我们使用四元数来描述AUV的姿态。为了控制姿态，我们实现了一个姿态控制器，该控制器是基于Chaturvedi等人[8]的工作推导出来的：

$$\omega_{\text{ref}}(\mathbf{R}, \mathbf{R}_d) = \mathbf{K}_\Omega \sum_{i=1}^3 \mathbf{e}_i \times (\mathbf{R}_d^\top \mathbf{R}^\top \mathbf{e}_i). \quad (25)$$

姿态控制器基于旋转矩阵 $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ (当前方向) 和 $\mathbf{R}_d \in \mathbb{R}^{3 \times 3}$ (期望方向) 生成角速度指令 $\omega_{\text{ref}} \in \mathbb{R}^3$ 。这两个矩阵可以直接从当前和期望的四元数计算得出。 $[\mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3]$ 是单位矩阵。 $\mathbf{K}_\Omega$ 是比例增益矩阵，其元素在实际实验中选取，以使俯仰上升动作的超调量最小：

$$\mathbf{K}_\Omega = \text{diag}(0.15, 0.15, 0.15). \quad (26)$$

为了避免过大的角速度， ω_{ref} 被限制在 $\|\omega_{\text{ref}}\|_{\text{max}} < 0.2 \text{ rad/s}$ 。

G. Depth Controller

为了保持AUV的深度稳定，实现了一个比例增益深度控制器：

$$\vec{v}_{\text{ref}} = \mathbf{K}_{\text{pos}} \mathbf{R}_B^I [[0, 0, z_{\text{ref}}]^\top - [0, 0, z]^\top], \quad (27)$$

其中 z_{ref} 是指令深度， z 是实际测量深度。 $\vec{v}_{\text{ref}} \in \mathbb{R}^3$ 是指令线速度， $\mathbf{R}_B^I$ 是从世界坐标系到机体坐标系的旋转矩阵。 $\mathbf{K}_\Omega$ 是比例增益矩阵，其元素在实际实验中选择，以实现最小超调：

$$\mathbf{K}_{\text{pos}} = \text{diag}(0.5, 0.5, 0.5). \quad (28)$$

最大线速度指令 $\vec{v}_{\text{ref}}$ 被限制在 $\pm[0.5 \text{ 米/秒}, 0.5 \text{ 米/秒}, 0.2 \text{ 米/秒}]$ 。

三、评估

A. Experimental Setup

表 I: *Cuttlefish* 规格 [4]

Specification	Value
Length / Width / Height	2.8 m / 2.0 m / 0.8 m
Weight	1200 kg
Inertial Navigation System	iXblue Phins Compact C3
DVL	Rowe SeaPilot
Pressure Sensor	Keller PAA-33x
Rated Driving Power	21.6 kW (8x 2.7 kW)

实验在德国不来梅DFKI RIC的海事测试池中进行¹。目标车辆是

¹<https://robotik.dfki-bremen.de/en/research/research-facilities-labs/maritime-infrastructure>

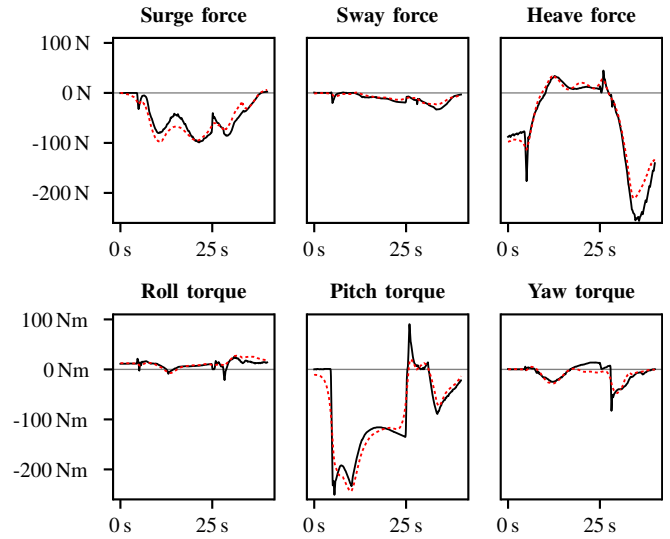


图 4: 模型识别测试 (虚线: 预测)

双臂干预无人潜航器 (AUV) *Cuttlefish* [4]，基于钢架结构建造，配备了13个用于电子设备的压力舱和基于PVC的泡沫，以使车辆具有正浮力。这两个机械臂在我们的实验中未使用，并且固定在车体上。为了控制重达1200公斤的车辆，安装了八个2.7千瓦轮缘驱动推进器（四个垂直推进器和四个水平推进器）。这些推进器由Wittenstein Cyber Motor GmbH专为*Cuttlefish*开发，设计用于精确控制，即使在非常低的转速下也是如此。推进器加速的实测时间常数为 $T_{\text{thr}} = 0.672 \text{ s}$ 。为了估计姿态、速度和线性加速度，我们使用iXblue Phins Compact C3惯性导航系统。Phins C3基于光纤陀螺仪，提供高精度的旋转速率测量，分辨率为0.001°/s。此外，Phins C3能够估计滚转和俯仰方向的姿态。

TABLE II: Model Identification RMSE

DOF	Linear Model	Quadratic Model
Surge	11.31 N	9.38 N
Sway	6.62 N	5.01 N
Heave	16.92 N	18.94 N
Roll	6.39 Nm	6.86 Nm
Pitch	17.91 Nm	15.83 Nm
Yaw	13.74 Nm	13.40 Nm

B. Model Identification

In order to identify the motion model, we conducted a series of test drives where we recorded 57 minutes of training data in total. During the first test drive, we manually carried out random motions with the AUV, incorporating coupled maneuvers in different axes and orientations. We then identified a preliminary motion model, which was used in a second test drive. During the second test drive we did some pitch-up maneuvers using the FBL-based controller, in order to capture some training data representing the actual task we are aiming for. After collecting all training data, we identified the final motion model and did a test drive to collect a two-minute dataset solely used for testing the model identification performance, where we did three successive pitch-up maneuvers. The root mean square error of the predicted wrench is shown in Table II for a linear drag model and for a quadratic drag model. Figure 4 shows one full period of the test sequence, where the vehicle transitions into a upright pose and back into a horizontal pose. The dashed curve represents the predicted force/torque using the quadratic damping model, while the solid curve represents the actual force/torque. It can be seen that the model captures the coarse motion of the AUV accurately. However, for example the yaw prediction has an inaccuracy at 20s, and the peak in the heave DOF is not correctly predicted at 10s.

C. Experimental Results

During our real-world experiments we compare the INDI-based controller to a linear-drag and a quadratic-drag feedback-linearized controller.

1) *Pitch-up maneuver*: In our first experiment we evaluate the transition from horizontal into upright pose by measuring the step response from 0° pitch to -90° pitch for 30 seconds. Subsequently, we perform a transition back to the stable horizontal pose, again for 30 seconds. The whole process is repeated for three cycles. Figure 5 shows the resulting plots. The first row shows the pitch velocity for each controller, and the second row shows the actual pitch torques, as calculated from the thruster outputs. The third row shows the actual pitch angles, while the desired pitch angle is outlined by a dashed curve. Looking at the plots we notice that the performance of all three controllers perform similarly, with a slight overshoot at the upright phase and a larger overshoot at the stabilization phase. The mean overshoot during the pitch-up maneuver is 1.36° for INDI, 1.33° for linear FBL, and 1.59° for quadratic FBL. During stabilization, the mean overshoot is 1.71° for INDI, 2.63° for linear FBL, and 2.10° for quadratic FBL. The mean energy consumption

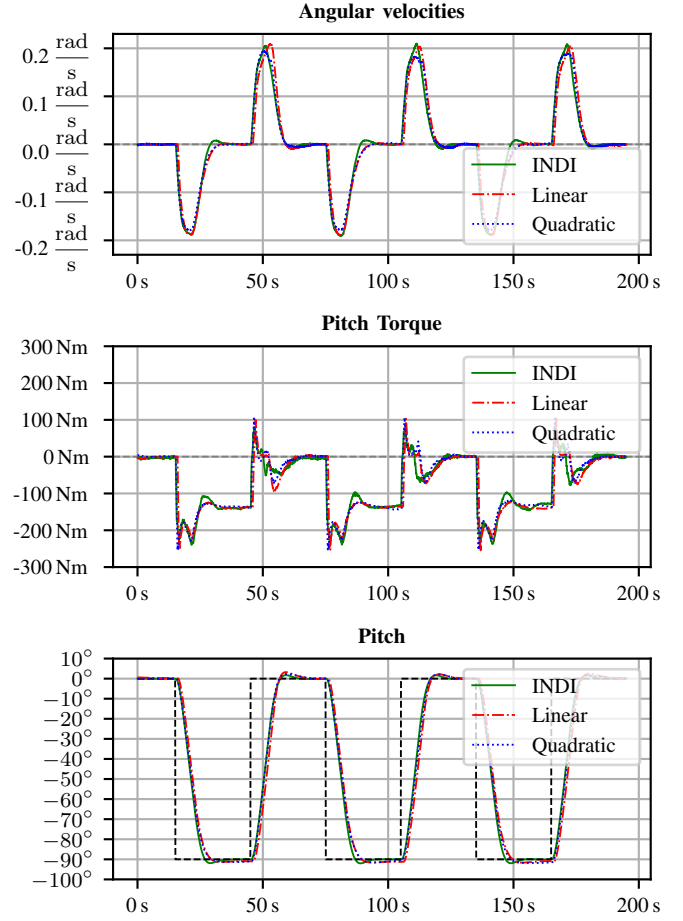


Fig. 5: Three pitch-up maneuvers

of INDI is 33.24 Wh per cycle, for linear FBL it is 33.84 Wh, and for quadratic FBL it is 34.19 Wh. We notice that both feedback-linearized controllers result in a steady-state orientation error in upright pose. To study the steady-state error in more detail, we maintain the upright pose for 300 seconds and then measure the error. For INDI, we measure an error of 0.0829° . For the feedback-linearized controllers we measure an error of 1.3459° for the linear drag model, and 1.6897° for the quadratic drag model. The reason INDI has a smaller steady-state error is that it has an incremental control-law, which is eliminating the nonlinear effects successively. Table III shows the RMS tracking errors for the linear and angular velocities. According to the data, INDI keeps the roll axis more stable than the model-based controllers, which indicates that INDI captures the coupled dynamics better. A surprising result is that the tracking performance of the quadratic drag model is worse than that of the linear drag model. One possible explanation could be that the velocities of our system are considerably low, which makes the quadratic drag terms hard to identify correctly.

2) *Station Keeping*: To evaluate the station keeping performance, we measure the positional drift while maintaining an upright pose for 300 seconds. The drift is plotted in Figure 6 for INDI and FBL. It can clearly be noticed that the feedback-linearized controllers have much higher drift. One

表 II: 模型识别均方根误差

DOF	Linear Model	Quadratic Model
Surge	11.31 N	9.38 N
Sway	6.62 N	5.01 N
Heave	16.92 N	18.94 N
Roll	6.39 Nm	6.86 Nm
Pitch	17.91 Nm	15.83 Nm
Yaw	13.74 Nm	13.40 Nm

B. Model Identification

为了识别运动模型,我们进行了系列测试驾驶,总共记录了57分钟的训练数据。在第一次测试驾驶中,我们手动操作AUV进行随机运动,包括在不同轴和方向上的耦合机动。然后我们识别出了初步运动模型,该模型用于第二次测试驾驶。在第二次测试驾驶中,我们使用基于FBL的控制器进行了部分俯仰上升机动,以获取一些代表我们实际任务的训练数据。收集完所有训练数据后,我们识别出了最终的运动模型,并进行了测试驾驶,以收集一个仅用于测试模型识别性能的两分钟数据集,其中进行了三次连续的俯仰上升机动。预测力矩的均方根误差在表II中显示,分别针对线性阻力模型和二次阻力模型。图4显示了测试序列的一个完整周期,其中车辆进入上升状态

C. Experimental Results

在我们的真实环境实验中,我们将基于INDI的控制器与线性阻力和二次阻力反馈线性化控制器进行比较。

1) *Pitch-up maneuver*: 在我们的第一个实验中,我们通过测量从0°俯仰到90°俯仰的阶跃响应来评估从水平姿态到竖直姿态的过渡,持续时间为30秒。随后,我们进行回到稳定水平姿态的过渡,同样持续30秒。整个过程重复三次循环。图5显示了结果图。第一行显示每个控制器的俯仰速度,第二行显示由推进器输出计算得到的实际俯仰力矩。第三行显示实际俯仰角,而期望的俯仰角则由虚线曲线表示。从图中可以看出,三个控制器的性能表现相似,在竖直阶段略有超调,在稳定阶段超调较大。俯仰上升过程中平均超调为INDI控制器的1.36°,线性FBL控制器的1.33°,二次FBL控制器的1.59°。在稳定阶段,平均超调为INDI

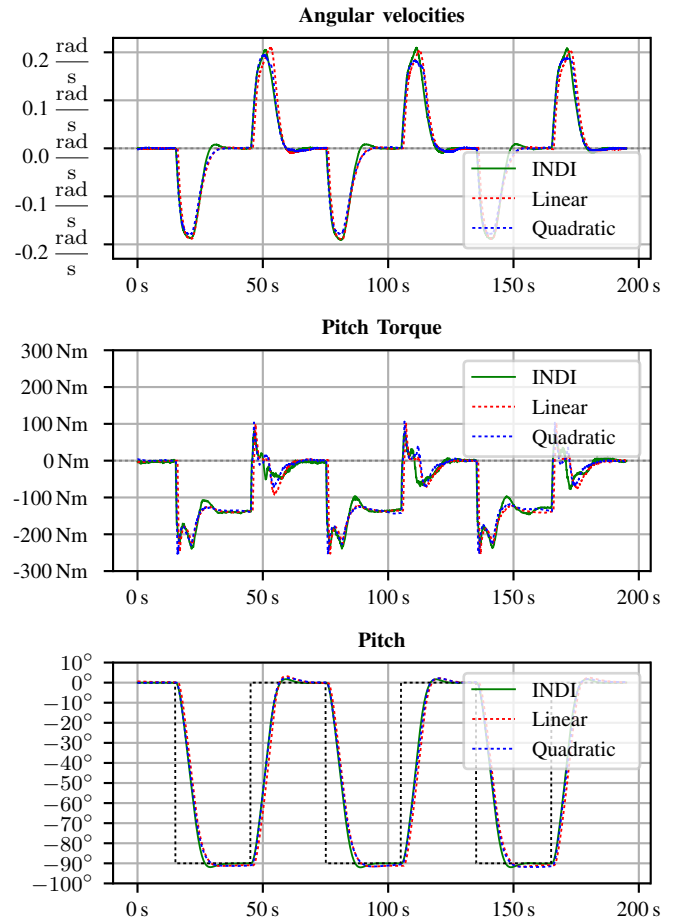


图5: 三次抬头机动

INDI 的每个周期能耗为 33.24 瓦时,线性 FBL 为 33.84 瓦时,二次 FBL 为 34.19 瓦时。我们注意到,两种反馈线性化控制器在直立姿态下都会产生稳态姿态误差。为了更详细地研究稳态误差,我们保持直立姿态 300 秒,然后测量误差。对于 INDI,我们测得误差为 0.0829°。对于反馈线性化控制器,我们测得线性阻力模型误差为 1.3459°,二次阻力模型误差为 1.6897°。INDI 的稳态误差较小的原因是其具有增量控制律,可以连续消除非线性效应。表 III 显示了线速度和角速度的均方根跟踪误差。根据数据,INDI 能比基于模型的控制器更稳定地保持滚转轴,这表明 INDI 更好地捕捉了耦合动力学。一个令人惊讶的结果是二次阻力模型的跟踪性能

2) *Station Keeping*: 为了评估定点保持性能,我们在保持直立姿势300秒的过程中测量位置漂移。图6中绘制了INDI和FBL的漂移情况。可以明显注意到,反馈线性化控制器的漂移要高得多。一个

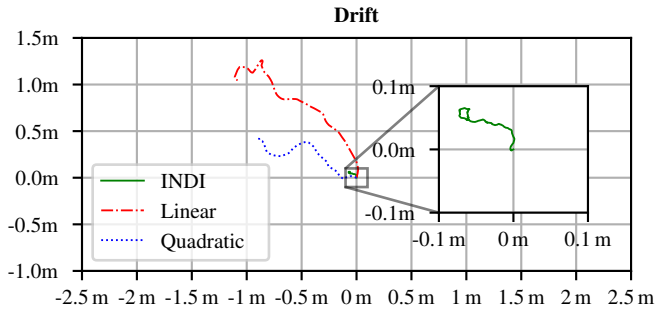


Fig. 6: Drift after 300 s in x and y world coordinates

TABLE III: Tracking Error (RMSE)

	INDI	Linear FBL	Quadratic FBL
Surge	0.39 mm/s	0.55 mm/s	1.04 mm/s
Sway	0.12 mm/s	0.34 mm/s	0.64 mm/s
Heave	0.73 mm/s	0.69 mm/s	1.11 mm/s
Roll	0.022 °/s	0.043 °/s	0.063 °/s
Pitch	0.328 °/s	0.321 °/s	0.317 °/s
Yaw	0.053 °/s	0.051 °/s	0.108 °/s

explanation is that feedback linearization tries to compensate the damping terms. However, if the modeled drag is higher than the actual drag, the linearization loop applies too much force, resulting in instabilities. A detailed overview of the error velocities is given in Table IV. During station keeping, we could measure a continuous power consumption of 2231 Watts using INDI, 2246 Watts using linear-drag FBL, and 2274 Watts using quadratic-drag FBL.

IV. CONCLUSION

In this paper, we conducted a first study on using incremental nonlinear dynamic inversion for attitude control of a 6-DOF underwater vehicle. Our results show that INDI clearly outperforms the model-based controller in stabilizing the upright pose with minimum drift. Keeping the drift of the AUV minimal is crucial for our intended use in intervention tasks. INDI results in a much lower steady-state error than the model-based controller, while keeping the roll axis much more stable during transition phase. Our findings are as expected, since the installed PHINS Compact C3 is a state-of-the-art military grade INS. It is possible that INDI might not have outperformed the model-based controller when less accurate sensor hardware or more advanced models had been used. However, modeling the dynamic behavior of marine vehicles is difficult, and not all hydrodynamic effects can be modeled exactly. For our sensor-based approach, only a 6×6 mass-inertia matrix together with an actuation model is needed, reducing the modeling effort significantly. This also comes with the advantage that altered dynamics have less impact on controller performance.

The results of this work can be seen as a basis for further studying INDI on marine vehicles. According to [14], INDI is also capable of adapting to actuator failure to a certain extent, which has not been studied in this work. For this reason, the authors plan to conduct upcoming research on Fault-Tolerant Control using INDI.

TABLE IV: Station Keeping Error (RMSE)

	INDI	Linear FBL	Quadratic FBL
Surge	0.128 mm/s	0.449 mm/s	0.279 mm/s
Sway	0.016 mm/s	0.171 mm/s	0.088 mm/s
Heave	0.017 mm/s	0.191 mm/s	0.117 mm/s
Roll	0.003 °/s	0.015 °/s	0.015 °/s
Pitch	0.007 °/s	0.066 °/s	0.049 °/s
Yaw	0.007 °/s	0.027 °/s	0.029 °/s

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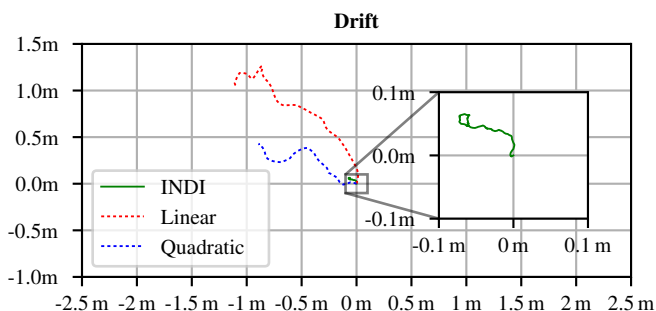


图6：在世界坐标系 x 和 y 中 300 秒后的漂移

表 III：跟踪误差（均方根误差）

	INDI	Linear FBL	Quadratic FBL
Surge	0.39 mm/s	0.55 mm/s	1.04 mm/s
Sway	0.12 mm/s	0.34 mm/s	0.64 mm/s
Heave	0.73 mm/s	0.69 mm/s	1.11 mm/s
Roll	0.022 °/s	0.043 °/s	0.063 °/s
Pitch	0.328 °/s	0.321 °/s	0.317 °/s
Yaw	0.053 °/s	0.051 °/s	0.108 °/s

解释是反馈线性化试图补偿阻尼项。然而，如果建模的阻力高于实际阻力，线性化环路会施加过大的力，导致不稳定。表 IV 给出了误差速度的详细概览。在定点保持期间，我们使用 INDI 测得持续功耗为 2231 瓦，使用线性阻力 FBL 为 2246 瓦，使用二次阻力 FBL 为 2274 瓦。

四、结论

在本文中，我们首次研究了使用增量非线性动态反演（INDI）进行6自由度水下航行器姿态控制。我们的结果显示，INDI在稳定直立姿态并将漂移最小化方面明显优于基于模型的控制。在干预任务中，将AUV的漂移保持在最小范围内至关重要。与基于模型的控制相比，INDI在稳态误差方面更低，同时在过渡阶段保持滚转轴更加稳定。我们的发现符合预期，因为所安装的PHINS Compact C3是一款先进的军用级惯性导航系统。在使用精度较低的传感器硬件或更高级的模型时，INDI可能不会优于基于模型的控制。然而，海洋航行器的动态行为建模十分困难，并非所有水动力效应都能精确建模。对于我们的基于传感器的方法，仅需一个6×6的质量惯性矩阵以及一个驱动方式

这项工作的结果可以被视为进一步研究海洋车辆上INDI的基础。根据[14]，INDI在一定程度上也能够适应执行器故障，而这在本工作中尚未研究。为此，作者计划在未来的研究中使用INDI进行容错控制的研究。

表 IV：保持站位误差（均方根误差）

	INDI	Linear FBL	Quadratic FBL
Surge	0.128 mm/s	0.449 mm/s	0.279 mm/s
Sway	0.016 mm/s	0.171 mm/s	0.088 mm/s
Heave	0.017 mm/s	0.191 mm/s	0.117 mm/s
Roll	0.003 °/s	0.015 °/s	0.015 °/s
Pitch	0.007 °/s	0.066 °/s	0.049 °/s
Yaw	0.007 °/s	0.027 °/s	0.029 °/s

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